

Reproducibility in Research

Session Objectives

By the end of this session, you'll be able to:

- Identify issues with reproducibility in research, and best practices to make research reproducible.
- Discuss pros and cons of using Excel for spreadsheet data.
- Explain the difference between Excel files (.xlsx) and plain-text .csv files, and their role in reproducibility.
- Define scripting and the concept of literate programming.
- Connect the data science workflow and the use of R and RStudio with the research data lifecycle.

What is Reproducibility?

What is reproducibility?

- Research is reproducible when the **same data + the same methods = the same results.**
- Reproducibility is related to replicability, though replicability occurs when new data + the same methods = results that point in the same direction.

What is reproducibility?

		Data	
		Same	Different
Code & Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

What is reproducibility?

- Reproducibility in research can take different forms, including empirical/methodological, statistical, and computational.
- This program is going to focus on computational reproducibility, noting that this is one piece of a bigger picture.

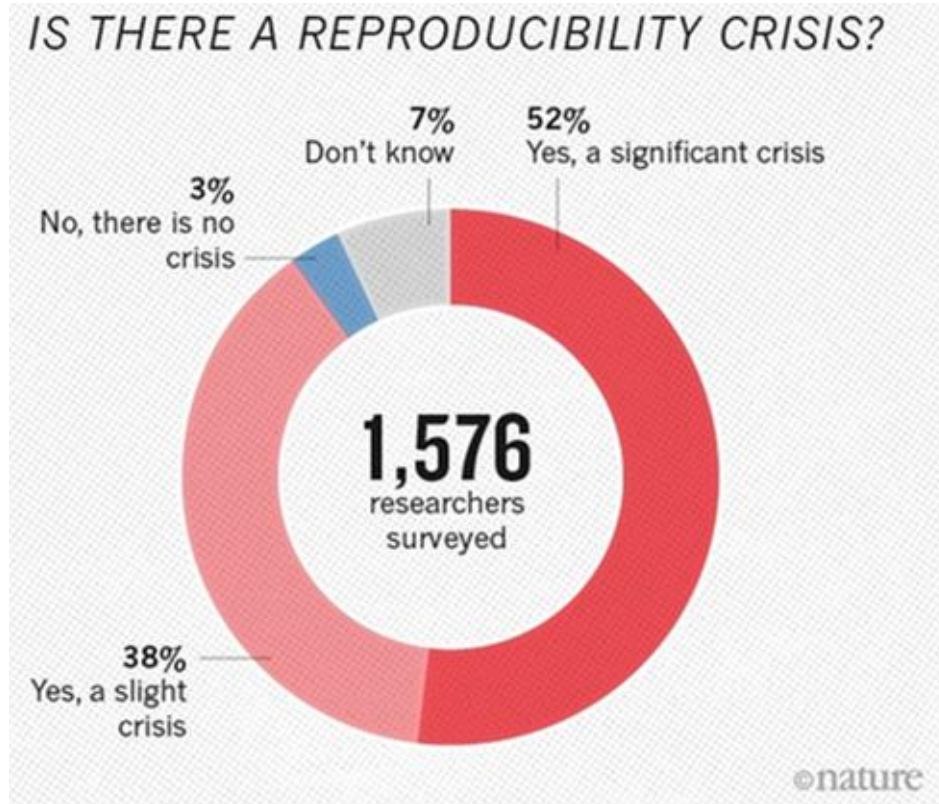
Computational reproducibility

For a project to be computationally reproducible, information about all assets used to test hypotheses and generate results must be available. These include, but are not limited to:

- Input data
- Code
- Software
- Computing environment and/or hardware

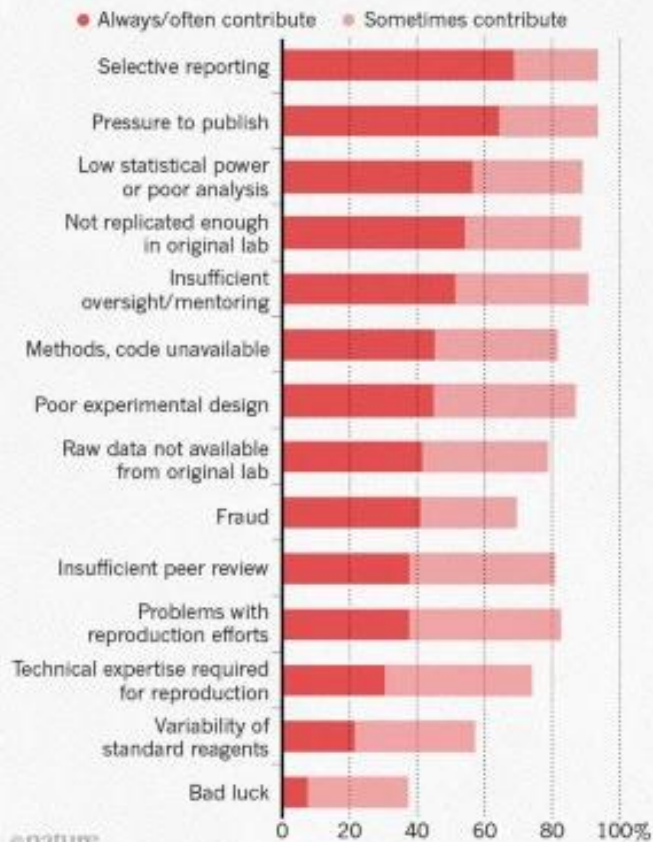
The Reproducibility Crisis

The reproducibility crisis



WHAT FACTORS CONTRIBUTE TO IRREPRODUCIBLE RESEARCH?

Many top-rated factors relate to intense competition and time pressure.





[eLife](#). 2021; 10: e67995.

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PMCID: PMC8651289

PMID: [34874008](https://pubmed.ncbi.nlm.nih.gov/34874008/)

Challenges for assessing replicability in preclinical cancer biology

We conducted the [Reproducibility Project: Cancer Biology](#) to investigate the replicability of preclinical research in cancer biology. The initial aim of the project was to repeat 193 experiments from 53 high-impact papers, using an approach in which the experimental protocols and plans for data analysis had to be peer reviewed and accepted for publication before experimental work could begin. However, the various barriers and challenges we encountered while designing and conducting the experiments meant that we were only able to repeat 50 experiments from 23 papers. Here we report these barriers and challenges. First,

Nuances of reproducibility

- Reproducibility is a spectrum.
- The more assets and information provided, the easier it is to reproduce the study.

Nuances of reproducibility

- **High / full reproducibility:** studies that provide the code, data, and full computational environment necessary to reproduce the results.
- **Medium / partial reproducibility:** studies that provide the code and data, but not the computational environment in which the code can be run.
- **Low / shallow reproducibility:** studies that provide algorithms and results.

Nuances of reproducibility

- **Sharing $\neq$ reproducible.**
- Simply sharing data and code does not make a project reproducible.
- Could another researcher, without ever speaking to you, open, understand, run, or reuse your data and code?

Nuances of reproducibility

- **Not all data are meant to be computationally reproducible.**
- Qualitative, humanities, or arts-based research are not exempt from good documentation practices just because they are less computationally reproducible than other projects.

Best practices

- **Good RDM is not about perfection.**
- Slip-ups happen, and the focus should be on doing the best you can.

Best practices

- Be proactive
- Document, document, document
- Be FAIR

Working with Microsoft Excel

Have you ever used Excel?

- What did you use it for?
- Did it do a good job? Why/why not?

Share your answers in the chat, or raise a hand!

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Excel: What it does well

- Intuitive point-and-click graphical user interface (GUI)
- Easily group and order datasets based on specific values
- Easily calculate things like sum, mean, and other basic statistics
- Create graphs and charts with a few clicks
- **It's easy and convenient**

Excel: What it doesn't do so well

- Any manual changes made to a document are subject to human errors, and it can be hard to track these changes and identify errors.
- Formulaic changes can be hidden in cells and are subject to “breaking”, making manipulations and analyses difficult to track and reproduce.
- Manipulations require manual processes for every file.

Excel: What it doesn't do so well

- The auto-formatting of cells can be very annoying and cause issues.
- Rows/columns can be hidden, causing issues with collaboration.
- Changing variables will overwrite original data, causing issues with provenance and reproducibility.
- **It's not very reproducible.**

What's in a Spreadsheet?

.xlsx files

- Excel spreadsheets save as a **.xlsx** file, which is a proprietary Microsoft format.

.xlsx files

- While it's natural to think of an Excel file as a single file, it is actually a zipped archive containing multiple files, including:
 - Worksheet data
 - Worksheet formatting
 - Charts and graphs
 - Pivot tables
 - Embedded objects
 - External links
 - Metadata

.csv files

- **.csv** files (Comma Separated Values) are tabular (spreadsheet) files, with each row representing a record and values within each row separated by commas (even though they can be presented in cells).

Converting Excel files to other formats

- .xlsx files can be converted to plain text formats, but there are factors to be aware of:
 - When converting .xlsx files to plain text, all formatting (bold, colour, etc.) will be lost.
 - You can only convert one Excel tab to a plain text spreadsheet at a time.
 - Formulas and tables will be lost.

Converting Excel files to other formats

Original value/format	What can happen	Example of change	Why it happens
01234	Leading zero lost	1234	Treated as a whole number
Résumé	Characters corrupted	Rsum / RÃ©sumÃ©	Encoding issues
31/12/2024	Date misread or changed	12/31/2024	Different regional settings for dates
=SUM(A1:A10)	Formula lost	55 (only value)	Plain text only keeps values
Hello	Formatting lost	Hello	Plain text doesn't store formatting
123456789012345	Rounded or scientific notation	1.23E+19 or rounded value	Precision limits

Example: Genetic data

In a survey looking at over 11,000 papers with Excel gene lists published between 2014 and 2020, more than 30% contained at least one gene name error caused by Excel's autoformatting.

This was due to gene names being converted to standard dates, internal data numbers (5 digits), and floating point numbers.

<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1008984>

Moving to Scripting

What is scripting?

- Writing lines of code, or "scripts", is creating a set of instructions that a coding program can perform.
- For the purpose of this program, these instructions are performed on a dataset (noting there can be broader usage).

Literate programming

- Code is a language that computers can understand, but it is not always easily read by humans.
- Literate programming is a method of providing a human-language, or narrative, explanation of how a script works in combination with the script itself, so that it can be accurately interpreted and reused.
- We've already discussed how to document our files and data, but as we move through the program we're also going to talk about implementing documentation in our code and literate coding principles.

When not to script

- Coding, like any language, takes time and regular practice to learn.
- As a beginner, creating scripted solutions to simple research tasks may take much longer than using an easily accessible tool such as Excel.
- While scripting is reproducible, it is not always necessary for small, one-off tasks.

When to script

Scripting can overcome the common issues with Excel that were described earlier:

- In scripting there are no 'manual' changes; everything from the data to the configuration can be tracked and checked.
- Scripts are saved as individual documents and provide a clear and ordered list of everything that was done.
- A single script can be used for multiple files, and individual lines of code can be taken from one script and altered slightly to perform the same task on different datasets.
- There is no graphical user interface to inadvertently obscure settings or hide data.
- Scripting can load and work on a copy of the source data, so original data isn't overwritten or corrupted.

Scripting in R

The R coding language

- R is an open-source programming language designed for data manipulation, statistical analysis, and visualization in research.
- It can be used on any operating system, it has comprehensive documentation, and it has an active user-support and training community.
- It is popular in academic settings because researchers from many disciplines can use it on data types that range from bioinformatics to demography.

The R coding language

Because R is open-source, it is easily reproducible, as there are no hidden algorithms or proprietary file formats. Many R users are willing to share their code openly, which means that novice users don't need to have perfect fluency in order to get started.

RStudio

While R is the programming language, RStudio is an Integrated Development Environment (IDE), which is a fancy term for a software application that makes working in R much more user-friendly.

RStudio

An analogy:

You are writing a document in Microsoft Word. In this case, the plain human language that you are typing would be R, and Microsoft Word, which is the software that allows you to view, format, add visuals, and save your work, is RStudio.

Choosing a coding language

- There are many factors that go into choosing a coding language, many of which depend on your needs and goals of data analysis.
- While we'll be focusing on R for this program, once you have learned one coding language, it isn't too big of a stretch to transfer those skills to another language.

Time to jump into R!

- In the next session we're going to start working with R/RStudio, so please open RStudio on your computer.
- **Any questions before we begin?**